Team Science: Examination of Factors Associated with Citation Metrics

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Author Note

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The authors made the following contributions. Kelly Cuccolo: Conceptualization, Writing - Original Draft Preparation, Writing - Review & Editing; Jordan Wagge: Conceptualization, Writing - Original Draft Preparation, Writing - Review & Editing; John Edlund: Conceptualization, Writing - Original Draft Preparation, Writing - Review & Editing; Erin M. Buchanan: Conceptualization, Writing - Original Draft Preparation, Writing - Review & Editing, Analysis, Visualization.

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Abstract

Specific disciplines have reported recent increases in scientific collaboration (i.e., co-authorship), which may be related to article metric benefits (i.e., increased citation rates). However, it is currently unclear what disciplines have seen increases in collaboration, and if this increase in collaboration is associated with increased citation rates. Additionally, little is known about how demographic characteristics of the authorship team (e.g., h-index, gender, number of participating institutions) may influence article metrics in tandem with team size. Understanding the impact of collaboration patterns across time and on article metrics is critical given the importance of these factors in hiring/tenure decisions, and the general growth of the scientific enterprise. We will use metadata provided by Elsevier's International Center for the Study of Research (ICSR) lab to investigate collaboration patterns across disciplines over time and to understand factors that influence article metrics. The broad research questions are: 1.) What is the pattern of team size distribution within various disciplines over time, and is team size related to article impact? 2.) What characteristics of team composition have the greatest influence on article metrics?

Keywords: keywords

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Team Science: Examination of Factors Associated with Citation Metrics

Scientific collaboration (i.e., co-authorship) has been increasing in many fields of study such as biomedical sciences1, medicine2, and psychology3. Notably, scientific collaboration is one factor that may positively influence article metrics (e.g., citation rates4,5), although the extent of this effect may vary by field (e.g., physics6). Within scientific collaboration, additional characteristics of the authorship team may exert influence on these metrics. Authorship team diversity is a key factor one may consider when looking at the impact of collaboration on article metrics. Author diversity, broadly speaking, has been associated with positive outcomes such as publication in high impact factor journals, rates of citations, and funding7,8. It is important to note that author diversity can be conceptualized along a continuum of factors such as: gender, physical location (e.g. United States, Brazil), institutional affiliation, discipline, and rank (e.g., associate faculty). Evidence suggests an impact of international collaboration, rank, and gender on article metrics in fields such as sport science research9, physics6, and psychology10, respectively,but a gap exists in which the cumulation of these factors over time and across disciplines has not yet been addressed. As such, this project aims to understand: 1a.) What is the pattern of team size distribution within various disciplines over time, 1b.) is team size related to article impact?, and 2.) What characteristics of team composition have the greatest influence on article metrics?

Question 1. What is the pattern of team size distribution within various disciplines over time, and is team size related to article impact?

Science is becoming increasingly collaborative; evidence for this trend includes an increase in co-authored peer-reviewed publications (compared to single-authored)5 as well as initiatives, calls, and funding opportunities for collaborative research from both private and public/government sectors. Further, some argue that scientific collaboration and large interdisciplinary/international teams are needed to effectively address modern societal

questions and problems11. Indeed, team size (i.e., number of co-authors) has been increasing in recent years1 across various disciplines including science and engineering, social sciences, and arts and humanities5. While some research suggests an overall trend towards collaboration5 these findings do not appear to apply to all disciplines equally.

Within the natural/physical sciences, subdisciplines vary in their collaborative growth. For instance, average team size in physics has reportedly doubled over the last 45 years, whereas for mathematics, the growth appears slower - with the average team size rising from 1.22 to 1.845. Grossman12 corroborates this effect, noting that for mathematics the average number of collaborators has grown since the 1940s with 1 in 8 papers having 2+ authors. On the other hand, a review of four major musculoskeletal science journals13 noted approximately a doubling in the number of co-authors on a publication (3.67–7.3) over the last 30 years. Likewise, biology has noted decreases in single-authored papers across specific subareas (i.e., Ecology, Genetics, Zoology and Botany14).

Among the social sciences, a pattern of increasingly collaborative authorship emerges. For example, sociology has reported increases in both collaboration and the average number of authors per publication during the period from 1935 to 200515. Specific publication outlets in psychology (e.g., Journal of Applied Psychology16) have seen increases in co-authored and decreases in solo-authored papers. Specific subfields of psychology have also noted similar changes in authorship trends (e.g., industrial-organizational psychology17). Various branches of business, such as economics18 and finance19 have also noted increases in co-authorship.

Practically speaking, collaboration might offer important benefits to authors, as number of authors appears to be associated with article impact (i.e., citation rates). Among publications indexed in the annual volumes of the Web of Knowledge (Thomson-Reuters) in 2007, single author papers received an average of 2.91 citations while multi-author papers exhibited an average of 5.3920; this difference was consistent

when examining citation averages from 1998 to 2007. Further probing this relationship it appears that the number of citations increases with the number of authors, up to a point (approx. 20 authors) after which the relationship becomes inconsistent. As such, collaboration may be rewarded by increased impact.

While most of the available research points to a positive relationship between collaboration and citations, the available research tends to examine manuscripts published in a narrow time frame (i.e., ten years21), and often looks at fields in isolation, specific journals17, and/or specific geographical regions22. As such, it is unclear how various disciplines have embraced collaboration over time and whether there has been any associated impact.

Question 2. What characteristics of team composition have the greatest influence on article metrics?

Although several disciplines report increases in co-authorship over the past several decades, it is important to ask the question about who is publishing, and with whom. Generally speaking, despite increases in women's representation in academia, women still are underrepresented when it comes to publishing co- and solo-authored papers23. In addition to authoring a smaller proportion of published work, research suggests women are also less likely to hold prominent authorship positions24,25. For example, women are underrepresented in regard to publishing articles in specific areas of medicine26,27, STEM fields (e.g., mathematics28), and social sciences28,29, although the extent of these differences varies by subfield (e.g., computational biology, as compared to biology in general29). Furthermore, women are less likely to receive credit for their contributions to empirical/scholarly written work30. Interestingly, women are less likely to collaborate with previous co-authors compared to men31; although it is not clear why this is the case, it could have important implications for their careers as novel teams tend to produce higher-impact work32.

Compounding this lack of representation is a body of research suggesting that women's contributions to science via peer-reviewed publications are often perceived to have lesser value than publications authored by men. After controlling for year of publication, publication outlet, focus, theoretical framework, methodology, tenure status, and institutional affiliation, women were still cited less than men33. Relevant to this manuscript is the body of research examining the association between gender and citation rates. Previous work has noted that in specific fields, men are less likely to cite women's work and work authored by women typically receives fewer citations overall34,35. This difference in likelihood of accumulating citations could influence a researchers' impact36, which is often considered important for career advances.

One objective measure of a scholar's output and impact is the "h-index"37. The h-index combines the number of articles published with the number of citations received for those articles. This approach attempts to balance high-impact articles with the number of articles published. Specifically, the h-score is the number of articles (X) that receive X citations. The h-index is one of the most widely used and accepted measures of a scholar's impact38. Fields such as psychology have observed gender differences pertaining to h-indices which favor men. Women professors have a significantly lower h-index than men, with a difference of approximately four points. After controlling for the year of first publication, gender shares approximately 2.6% of the variance with h-index10. This effect was examined relative to mean salaries for full professors, and practically, translated to an astounding effect of a $13,100 a year difference between men and women10.

It is important to note that there is evidence of increasing equity between genders in certain fields; in international relations, for example, the gap between men and women's manuscript submissions and publications are narrowing and there appears to be no significant relationship between gender and citation count39. Notable, is that the calculation of h-indices may be particularly biased by self-citations and years without

publication. Higher rates of citations received by men were partially explained by high rates of self-citation which altered scores up to 14 extra h-index points. Women also engaged in self-citation, although less so, and only impacting their h-index by up to 5 points34. Although there are alternatives to the h-index that account for self-citations (e.g. the b-index40), the h-index is popular and generally accepted as a robust bibliometric tool.

Given the arguments regarding the necessity of scientific collaboration and large international teams to address modern societal questions and problems11 it may be intuitive that institutional and geographical diversity is on the rise. One source notes an increase in articles that have the presence of both a U.S. author and at least one international co-author for psychology41. Similar increases in the fraction of internationally co-authored papers (i.e., ratio of co-authored articles with international institutions to the total number of co-authored articles with domestic and international institutions) were noted in other scientific disciplines such as mathematics, biology, medicine, and engineering42. However, while it appears most fields are increasing international engagement, applied fields (especially medical sciences) have higher rates of growth43.

Collaboration involving those outside of one's home institution and country may significantly impact article metrics, which may be one reason for the increasing rates of international collaboration. For instance, for UK institutions indexed in the Science Citation Index between 1981 and 1994, collaboration within one's home university or country resulted in a small increase in average impact (0.75 citations) whereas collaboration outside of one's home country resulted in a more moderate increase (1.6 citations)44. Indeed, both well-established and young institutions benefit from international collaboration as evidenced by a citation per paper increase when the authorship team is international45. This benefit is corroborated by the observation that international co-authorship publications are over-represented in the ESI global top 10 % highly cited publications relative to all journal publications from the same institution45.

Indeed, article impact in terms of citations per paper seem to be greatest for those manuscripts with multiple domestic and international collaborators, and smallest for those with no institutional collaboration.

However, this trend may not be consistent across fields, journals, and/or countries, and warrants further investigation (e.g., physics6,46). For example, international collaboration generally increases citation rates in the field of sports science but the extent of this effect varies by which countries are collaborating9. Collaboration between the United States and Canada for instance, produced a mean citation rate greater than the domestic mean observed citation rate of each country respectively, whereas collaboration between the United Kingdom and Greece produced a mean citation rate that was lower than the domestic mean observed citation rate of each country respectively. Furthermore, when examining the benefits of international collaboration on citation count for american authors publishing in the top three interdisciplinary journals (Proceedings of the National Academy of Sciences [PNAS], Nature, and Science), there is an increased citation rate for international collaboration when compared to American-only authored papers46. It is important to note that this effect was the result of a significant increase in citation rate for international papers published in PNAS only. When examining each journal independently, no benefit of international collaboration was found for Science or Nature. Given the accelerated growth of international collaboration across various fields over time43,47, it is increasingly important to understand the influence of this diversity on article metrics.

The majority of work examining team characteristics and article metrics have focused on a specific field6, and/or a limited/restricted number of team characteristics (e.g., size5). Empirical data comparing the impact of diverse/multiple team characteristics on article metrics within a given field is limited2,6,9,12,14. Along a similar vein, empirical data comparing the impact of team characteristics on article metrics across fields is also limited (for notable exceptions, see Larivière et al4 or Wuchty et al5). Comparing outcomes within

and across fields can allow for an increased understanding of these fields' scientific culture. Practically, understanding factors associated with an article's impact can be important for researchers' outreach, as well as planning for the job market and tenure review.

Method

Ethics Information. All of the data for this manuscript is drawn from peer reviewed journals. All of the extracted meta-data from the individual publications is de-identified in the database. As such, per US Federal Guidelines, this research is not subject to human ethics review boards (§46.101, (b) (4)).

Design. This study is an archival data investigation into scholarly publication records using Elsevier's curated database Scopus. This study will be conducted through access to Elsevier's International Center for the Study of Research Lab (ICSR; https://www.elsevier.com/icsr). The ICSR is designed to support researchers' in their empirical study of meta-science on large datasets using the Databricks platform running Apache Spark architecture for big-data analyses. Through ICSR, researchers can obtain access to full publication metadata from Scopus including article metrics, author(s), and affiliation information. Scopus is a global database on abstracts and citations, containing a wide representation of content including articles, books, and conference proceedings48. Scopus contains over 80 million entries from 1788 to present curated by a team of experts to ensure quality metadata and inclusion of appropriate publications. All variables extracted from this dataset are described below.

Variables. Article characteristics include the year of publication, number of authors (team size), and discipline. Discipline will be defined at the All Science Journal Classification (ASJC; https://service.elsevier.com/app/answers/detail/a_id/15181/supporthub/scopus/) using the subject area codes, resulting in 27 different discipline areas. Metric characteristics include citation count (number of times an article was cited within the current data

snapshot) and weighted citations. Field weighted citations indicates the number of citations in the last five years. We will check if the five-year count is excluded for newer articles, and if not, we will exclude all publications that have not been available for five years to ensure this metric is consistent.

Author characteristics include affiliation country, affiliation/institution, h-index, gender, and gender ratio. We will transform the affiliation country into a count variable of the number of unique countries. Similarly, we will transform the affiliation/institutions into a count of the number of different affiliations across authors. We will use the h-index of the first and last authors for each article. We will use h-index with and without self-citation. Last, Elsevier has noted that their inferred gender column has a 97% accuracy, and we will use inferred gender of the first and last authors for each article. From this variable, we will additionally calculate the gender ratio (number of inferred male / total authors) as a single variable for each article.

Update: We will use the larger subject areas (Physical, Life, Health, Social Sciences, and Multidisciplinary) due to constraints on the number of analyses and time taken to run all 27 areas on the Databricks server.

Sampling Plan. While Scopus includes articles from 1788 to present, we will limit our sample to journal articles published from 1970 until present. Scopus indicates that articles from 1970 to present include reference information, which will be required for calculation of several of the variables described above. Only journal articles from this time period will be included. No other exclusions will apply. We will include the date of analysis to indicate what data was available at the time analysis.

Update: We will sample from 25% of the Scopus dataset from 1970 until present. This size limitation is due to the amount of time and memory necessary to run all analyses including h-index calculation.

Power. Given the large size of the Scopus dataset, all analyses are expected to achieve the required 95% power. Analyses using the pwr library49 indicate that planned regressions have the ability to detect effects at R2 < .00001 at 95% power using a 95% confidence interval.

Analysis Plan.

Pilot Code. We were allowed to use a small subset of cases (1%) to create the analysis code for the proposed research questions. All code will be available through GitHub and the Open Science Framework.

Data Screening. Missing data will be excluded for the analyses in which the data is missing only. We will otherwise assume all data is accurate. In all analyses, we will examine the cell counts for each categorical variable in order to collapse small cell sizes (i.e., n < 5). No articles will be considered an outlier, as all publication data is expected to be part of the population.

In each relevant analysis, we will examine for multicollinearity of the IVs, as team size may positively correlate with count variables calculated in later research questions. If values are multicollinear, r > .90, we will note these correlations and only use one of the proposed variables. The size of the dataset implies the assumption of normality. In our analyses, a linear trend is the main focus of interest, and we will note any obvious non-linear trends. Last, we will note any violations of homoscedasticity. In order to feasibly examine these two assumptions (linearity, homoscedasticity), we will first run all analyses with the default (n >= 600,000) randomly selected data points and plot the residuals versus the predicted values for each analysis. We will summarize those findings in a supplemental file denoting potential deviations.

Update: This analysis was completed on the original pre-registration.

Decision Criteria. The large sample size will likely indicate all predictors significant at the traditional p value criterion. For individual predictors, we will use the

95% confidence intervals of the slope to indicate if the effect is non-zero (i.e., confidence interval does not include zero rounded to three decimals). The difference between effects will be determined by examining the confidence intervals of R2 to three decimal places. If they do not overlap, we will state that they are different.

Research Questions. All analyses will be performed using pyspark50 within the Databricks system. Visualization and presentation of analyses will be performed using R and ggplot251. We will report R2 and its 95% confidence interval for each model as an effect size using the MOTE package52. Research

Question 1a: What is the pattern of team size distribution within various disciplines over time? We will first calculate the mean, median, and interquartile range of team size by discipline. We will not explicitly compare discipline intercepts in a post hoc format with any form of test. Instead we will use visualization techniques to display the distribution and their interquartile range for readers to examine (i.e., a forest plot).

In order to determine if team sizes change overall across time, we will use a linear model of team size predicted by year of publication in which each article is only represented once, regardless of discipline. We will examine if the confidence interval of the slope parameter is different from zero to determine if the predictor was significant. Last, we will calculate each discipline's slope and confidence interval using a linear model of team size predicted by year of publication. Again, we will provide these slopes in a visualization for readers to examine with no official post hoc comparison. While a multilevel model would allow us to embed all disciplines within one model, no pyspark multilevel model implementation exists to model data at this scale.

Research Question 1b: Is team size related to article impact? For this research question, we will examine the impact of team size on article publication metrics for impact. Our hypothesis is that team size impacts article metrics (non-directional). We will run these analyses once for each dependent variable of citation count and weighted citation five

year count. We will calculate the mean, median, and interquartile range of each publication metric by discipline. We will visualize these values instead of comparing them directly.

To determine the impact of team size on publication metrics, we will use a linear model of publication metrics predicted by time (covariate) and team size. We will include time as a covariate, as we expect that older articles will potentially have more or different metrics because they have been around longer to cite. We will examine if the confidence interval of the team size slope parameter is different from zero. Last, we will calculate the slope for team size for each discipline separately. We will visualize these differences and the confidence intervals.

Research Question 2: What characteristics of team composition have the greatest influence on article metrics? For this research question, we will continue to use the model from the previous research question and add other article metrics to determine their contribution to the model. Our hypothesis is that each variable has an impact on the metrics variables (non-directional). Each author independent variable will be modeled separately and compared using confidence intervals as described above for which variable had the strongest impact on publication metrics.

Using the linear model from the previous research question, we will add each author independent variable (affiliation country count, affiliation/institution count, h-index, gender, and gender ratio) in separate models to calculate their slope. We will determine if these slopes were important by examining if their confidence interval was greater than zero. We have to examine the effects of each author metric separately. When examining first and/or last author information, the single author paper presents a unique problem because they are both first and last author. Therefore, these cannot be included in the same model. In order not to use the same variance twice, single author papers will be treated as first author only, and missing data for last author analyses. If these were included in the same model, all single author papers would be dropped, which would bias the results. We will

then compare each overall R2 to determine if their confidence intervals overlap. This analysis will indicate which author variable provides the most predictive information (since all other covariates are the same for each analysis, only the author variable is different). Last, we will calculate each author independent variable separated by discipline and then visualize those R2 and confidence intervals for each variable.

Data Availability

While we cannot provide access to the full dataset because it is only accessible through access granted by Elsevier, we will provide our analysis code via GitHub and the Open Science Framework. Persons interested in replicating or extending the current work can do so through using the ICSR lab Databricks platform. ICSR has stated to the authors of the paper that they will guarantee access to researchers who want to follow up or reproduce our results.

Code Availability

The code to run all analyses in the Databricks platform, as well as the code generated for this report can be found at:
https://osf.io/g4qnp/?view_only=d50da32d11cd47f889ce74cac954d2d7.

Update: We will sample from 25% of the Scopus dataset from 1970 until present. This size limitation is due to the amount of time and memory necessary to run all analyses including h-index calculation.

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Decision Criteria. The large sample size will likely indicate all predictors significant at the traditional p value criterion. For individual predictors, we will use the 95% confidence intervals of the slope to indicate if the effect is non-zero (lower bound greater than .001). The difference between effects will be determined by examining the confidence intervals of R^2 to three decimal places. If they do not overlap, we will state that they are different.

Research Questions. All analyses will be performed using *pyspark* (**pyspark?**) within the Databricks system. Visualization and presentation of analyses will be performed using *R* and *ggplot2* (Wickham, 2016). We will report R^2 and its 95% confidence interval for each model as an effect size using the *MOTE* package (**buchanan2019?**).

Research Question 1a: What is the pattern of team size distribution within various disciplines over time? **Update:** Discipline is defined as the five areas: Physical, Health, Life, Social, and Multidisciplinary. Note that papers can fit into multiple categories, and these analyses are therefore separated for each analysis.

We will first calculate the mean, median, and interquartile range of team size by

discipline. We will not explicitly compare discipline intercepts in a *post hoc* format with any form of test. Instead we will use visualization techniques to display the distribution and their interquartile range for readers to examine (i.e., a forest plot).

In order to determine if team sizes change overall across time, we will use a linear model of team size predicted by year of publication in which each article is only represented once, regardless of discipline. We will examine if the confidence interval of the slope parameter is different from zero to determine if the predictor was significant. Last, we will calculate each discipline's slope and confidence interval using a linear model of team size predicted by year of publication. Again, we will provide these slopes in a visualization for readers to examine with no official *post hoc* comparison. While a multilevel model would allow us to embed all disciplines within one model, no *pyspark* multilevel model implementation exists to model data at this scale.

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Research Question 2: What characteristics of team composition have the greatest influence on article metrics? For this research question, we will continue to use the model from the previous research question and add other article metrics to determine their contribution to the model. Our hypothesis is that each variable has an impact on the metrics variables (non-directional). Each author independent variable will be modeled separately and compared using confidence intervals as described above for which variable had the strongest impact on publication metrics.

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While we cannot provide access to the full dataset because it is only accessible through access granted by Elsevier, we will provide our analysis code via GitHub and the Open Science Framework. Persons interested in replicating or extending the current work

can do so through using the ICSR lab Databricks platform.

Code availability

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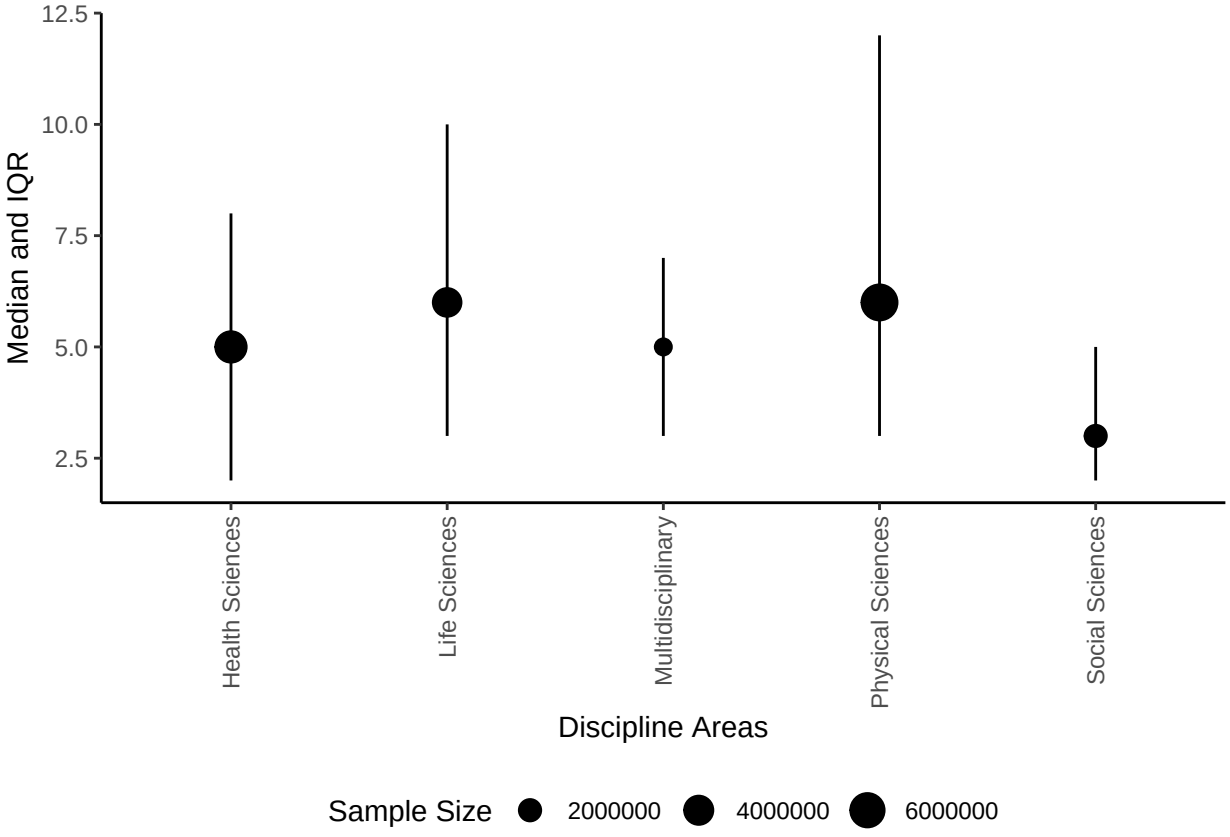
Results

Data Screening

all looked fine talk about the results in the supplement here.

Research Question 1a: What is the pattern of team size distribution within various disciplines over time?

Descriptive Statistics.

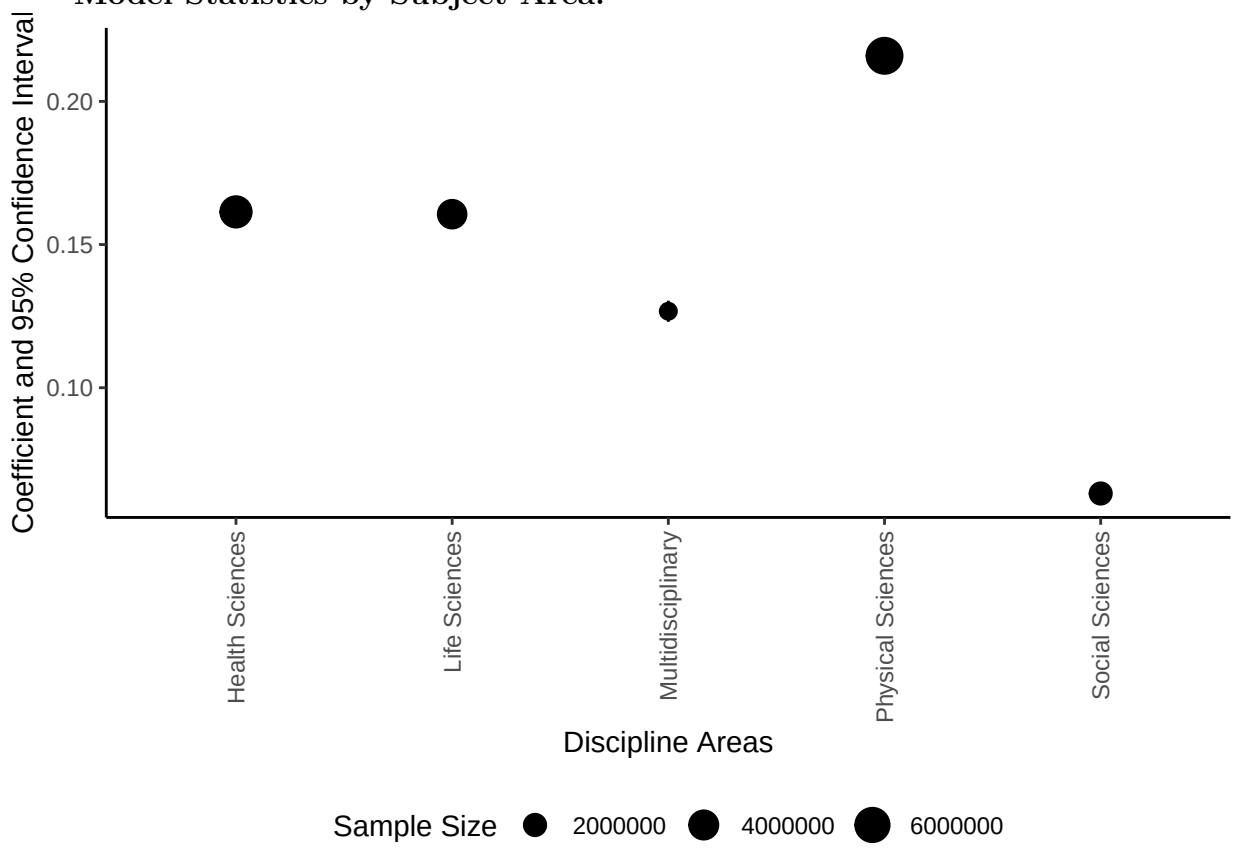


- We see median is approximately the same with social science as lower
- Physical has a larger top IQR top range

Model Statistics Overall. Does the confidence include zero? We will present $b = 0.08$, 95% CI [0.08, 0.08].

- X = team size, Y = year
- Team size is increasing across time

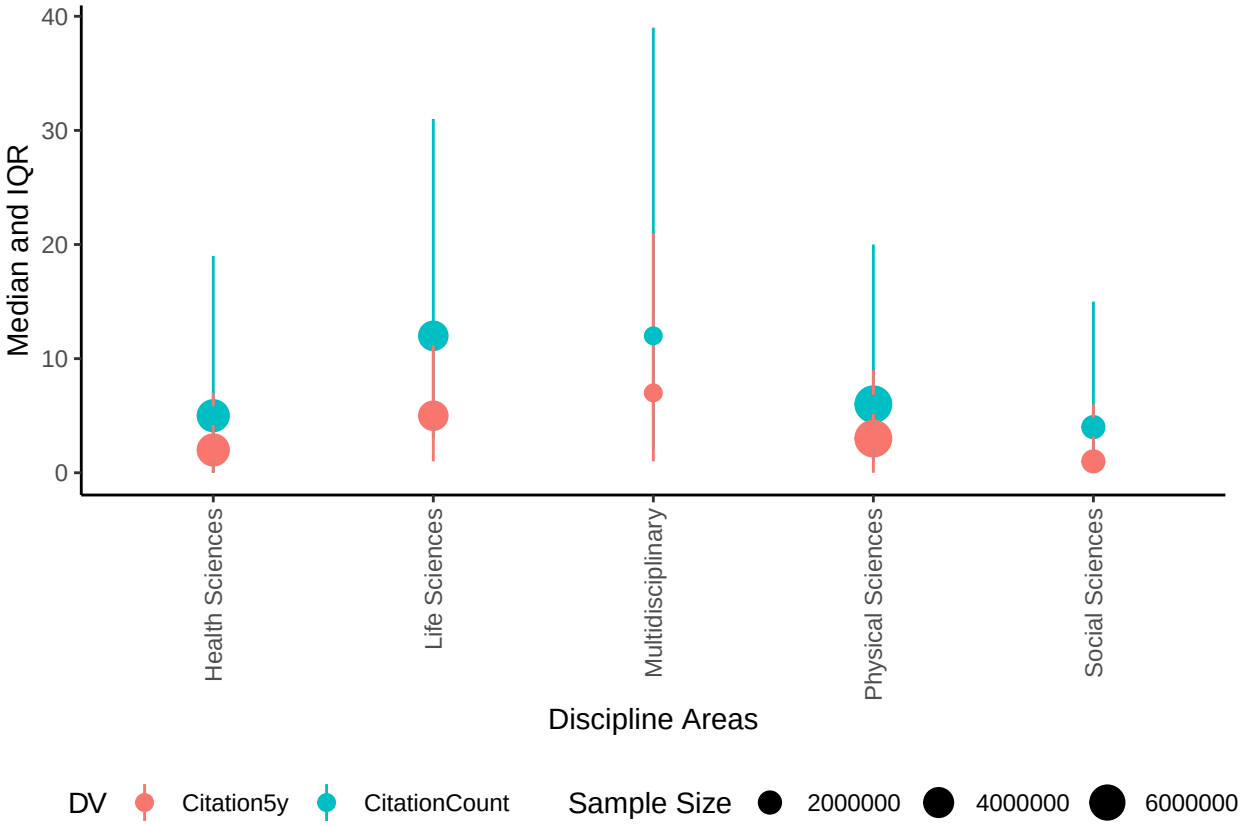
Model Statistics by Subject Area.



- All above zero
- Health and life similar
- Social lowest
- Physical highest

Research Question 1b: Is team size related to article impact?

Descriptive Statistics Citation Count.



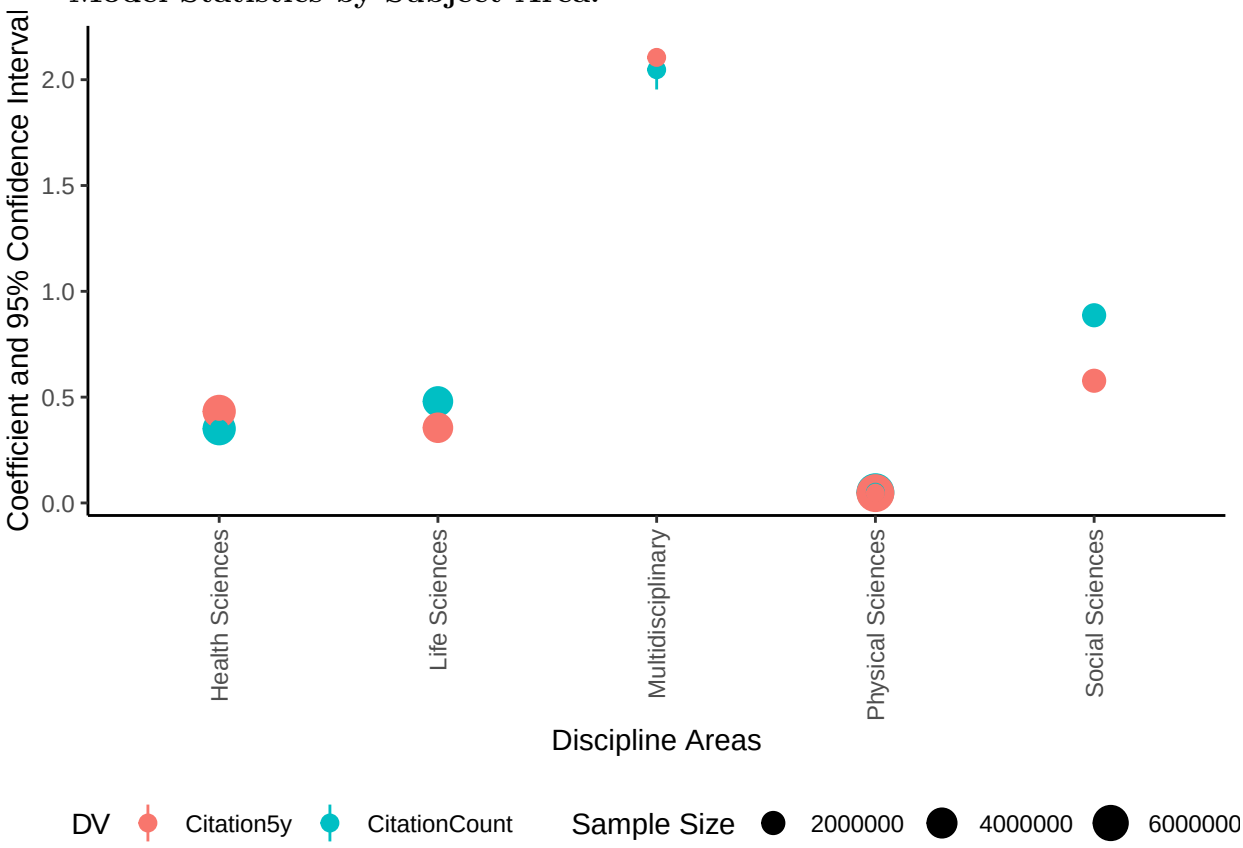
- Present mean, median, IQR?
- All fairly similar Multi has the highest range of citations

Model Statistics Overall. Does the confidence include zero? We will present:

- Citation Count = $b = 0.12$, 95% CI [0.12, 0.13]
- X = team size, Y = citation count, CV = Year
- Citation count is positively related to team size controlling for year
- controlling for team size, year is negatively related to citation count
- Citation 5ys = $b = 0.11$, 95% CI [0.11, 0.11]

- X = team size, Y = citation count, CV = Year
- Citation count is positively related to team size controlling for year
- Controlling for team size, year is positively related to citation count 5 year

Model Statistics by Subject Area.



- All positive, so citation count increases with team size
- Least = physical
- Most = multidisciplinary
- Most all of them have the negative-positive backwards relationship with Year except Multidisciplinary (both negative, so as year goes up citations go down)

Research Question 2: What characteristics of team composition have the greatest influence on article metrics?

Model Statistics Overall.

```
## Warning in pf(limit, nu, df2, ncp = LAMBDA.U): full precision may not have been
## achieved in 'pnbeta'

## Warning in pf(limit, nu, df2, ncp = LAMBDA.U): convergence failed in 'pnbeta'

## Warning in pf(limit, nu, df2, ncp = LAMBDA.U): full precision may not have been
## achieved in 'pnbeta'

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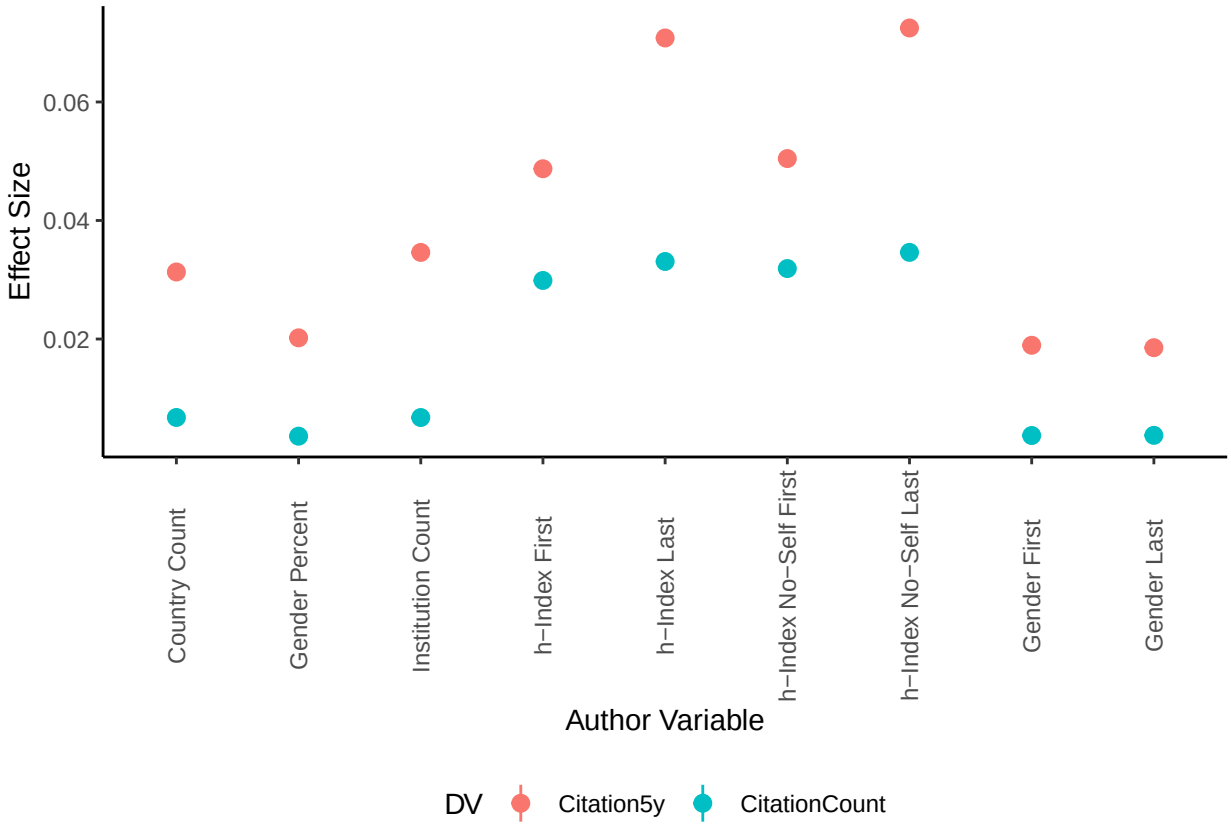
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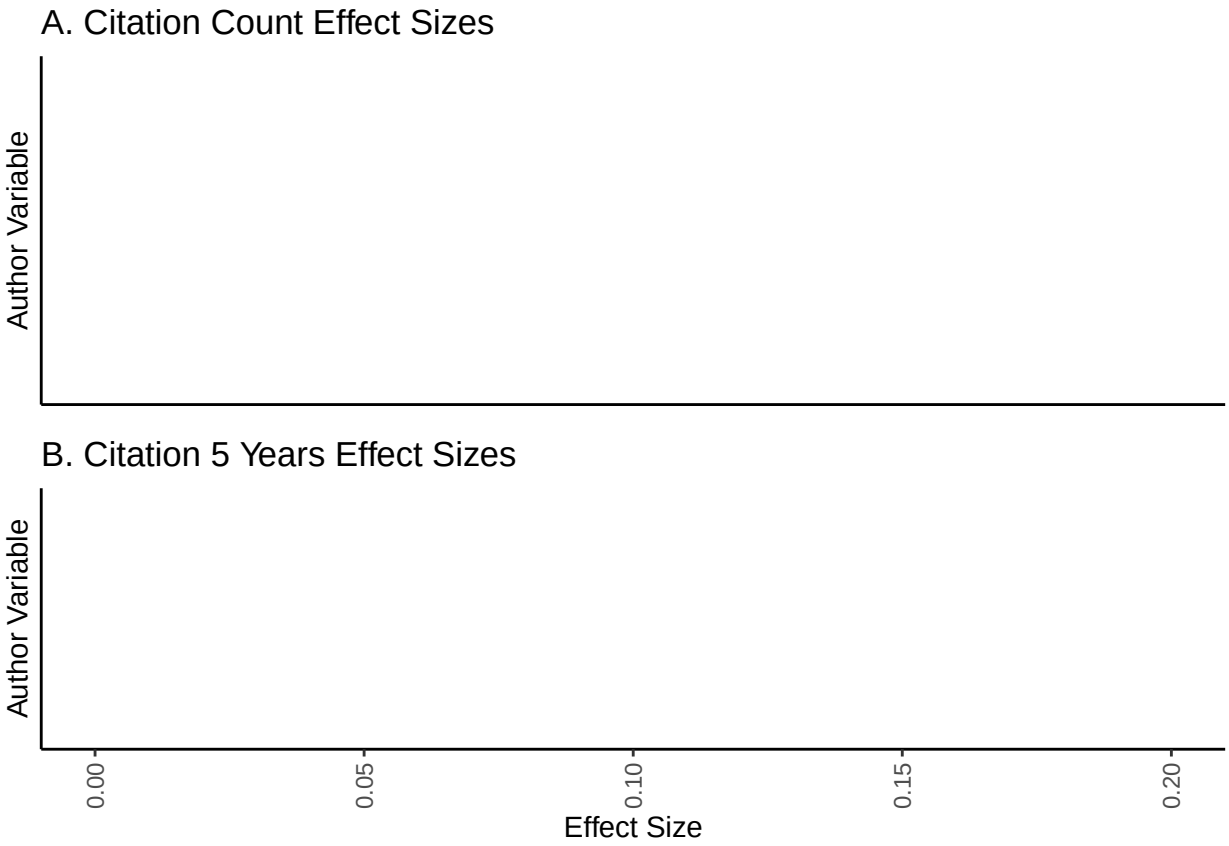
## Warning in pf(limit, nu, df2, ncp = LAMBDA.U): convergence failed in 'pnbeta'
```

- All values are "significant" using our rules
- Country count: as number of countries included increases, the number of citations increases
- Inst count: as the number of institutions included increases, the number of citations increases
- Gender percent (percent male): as more males increase, citation count decreases
- Hindex: as h-index increases, the number of citations increases for both first and last author, but is more for first
- Hindex no self: same effect but you get more citations as the h-index increases without self citations
- IPG gender female versus male: as it goes from female to male, we see an increase in citations, more so for the first author over the last (basically dudes get a bump)
- Do we want to talk about if team size is wiped out due to the author variable?
- Year still has that set of negative effects

We shall make a picture to denote which are different using R^2 as the metric since they are all different scales.



Model Statistics by Subject Area. Here's the statistics by subject area, but this graph may be crazy, we shall see. Right now this data is made up because it's still running online for the temporary data.


A. Citation Count Effect Sizes
Author Variable
B. Citation 5 Years Effect Sizes
Author Variable
0.00
0.05
0.10
0.15
0.20
Effect Size

577

578

References

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer-Verlag New York. Retrieved from https://ggplot2.tidyverse.org